

RAPILIN 30
Insulin Aspart Mix 30/70 100 U/mL Suspension for Injection

PRODUCT NAME

Rapilin 30 Insulin Aspart Mix 30/70 100 U/mL Suspension for Injection

COMPOSITION

Each ml contains:

100 U of soluble insulin aspart / protamine-crystallised insulin aspart* in the ratio 30/70 (equivalent to 3.5 mg).

1 cartridge contains 3 ml equivalent to 300 U.

*Insulin aspart is produced in *Escherichia coli* by recombinant DNA technology.

Contains M-cresol 1.72 mg and Phenol 1.50 mg as preservative.

Excipient: Glycerin, Phenol, M-cresol, Zinc chloride, Sodium phosphate dibasic anhydrous, Sodium chloride, Protamine Sulfate, Hydrochloric acid, Sodium hydroxide, Water for Injections.

PRODUCT DESCRIPTION

The suspension is cloudy, white and aqueous suspension for injection in cartridge.

Rapilin 30 is a biosimilar medicinal product of NovoMix 30.

PHARMACODYNAMIC

Pharmacotherapeutic group: Drugs used in diabetes. Insulins and analogues for injection, intermediate- or long-acting combined with fast-acting. ATC code: A10AD05.

Insulin aspart mix 30/70 is a biphasic suspension of 30% soluble insulin aspart (rapid-acting human insulin analogue) and 70% protamine-crystallised insulin aspart (intermediate-acting human insulin analogue).

Unless otherwise specified, data presented in this package insert for Efficacy and Safety were referred from package insert of NovoMix 30.

Mechanism of action and pharmacodynamics effects

The blood glucose lowering effect of insulin aspart is due to the facilitated uptake of glucose following binding of insulin to receptors on muscle and fat cells and to simultaneous inhibition of glucose output from the liver.

When insulin aspart mix 30/70 is injected subcutaneously, the onset of action will occur within 10 to 20 minutes

of injection. The maximum effect is exerted between 1 and 4 hours after injection. The duration of action is up to 24 hours.

In a 3-month trial comparing insulin aspart mix 30/70 with biphasic human insulin 30 administration before breakfast and dinner in patients with type 1 and type 2 diabetes, insulin aspart mix 30/70 resulted in significantly lower postprandial blood glucose after both meals (breakfast and dinner).

A meta-analysis including nine trials in patients with type 1 and type 2 diabetes showed that compared to biphasic human insulin 30, administration of insulin aspart mix 30/70 before breakfast and dinner resulted in significantly better postprandial blood glucose control (average prandial glucose increments over breakfast, lunch and dinner). While fasting blood glucose was higher in patients treated with insulin aspart mix 30/70, the overall glycaemic control measured by glycosylated haemoglobin was similar.

In one study, 341 patients with type 2 diabetes were randomised to treatment with insulin aspart mix 30/70 either alone or in combination with metformin, or to metformin together with sulfonylurea. HbA_{1c} after 16 weeks of treatment did not differ between patients with insulin aspart mix 30/70 combined with metformin and patients with metformin plus sulfonylurea. In this trial, 57% of the patients had baseline HbA_{1c} above 9%; in these patients treatment with insulin aspart mix 30/70 in combination with metformin resulted in significantly lower HbA_{1c} than metformin in combination with sulfonylurea.

In one study, patients with type 2 diabetes, insufficiently controlled on oral hypoglycaemic agents alone, were randomised to treatment with twice daily insulin aspart mix 30/70 (117 patients) or once daily insulin glargine (116 patients). After 28 weeks of treatment following the dosing guideline, the mean reduction in HbA_{1c} was 2.8% with insulin aspart mix 30/70 (mean at baseline = 9.7%). With insulin aspart mix 30/70, 66% and 42% of the patients reached HbA_{1c} levels below 7% and 6.5%, respectively, and mean FPG was reduced by about 7 mmol/L (from 14 mmol/L at baseline to 7.1 mmol/L).

In patients with type 2 diabetes, a meta-analysis showed a reduced risk of overall nocturnal hypoglycaemic episodes and major hypoglycaemia with insulin aspart mix 30/70 compared to biphasic human insulin 30. The risk of overall daytime hypoglycaemic episodes was increased in patients treated with insulin aspart mix 30/70.

Paediatric population: A 16-week clinical trial comparing postprandial glycaemic control of meal-related insulin aspart mix 30/70 with meal-related human insulin/biphasic human insulin 30 and bedtime NPH insulin was performed in 167 patients aged 10 to 18 years. Mean HbA_{1c} remained similar to baseline throughout the trial in both treatment groups, and there was no difference in hypoglycaemia rate with insulin aspart mix 30/70 or biphasic human insulin 30.

In a smaller (54 patients) and younger (age range 6 to 12 years) population, treated in a double-blind, cross-over trial (12 weeks on each treatment), the rate of hypoglycaemic episodes and the postprandial glucose increase was significantly lower with insulin aspart mix 30/70 compared to biphasic human insulin 30. Final HbA_{1c} was significantly lower in the biphasic human insulin 30 treated group compared with insulin aspart mix 30/70.

Elderly: The pharmacodynamic properties of insulin aspart mix 30/70 have not been investigated in the elderly. However, a randomised, double-blind, cross-over PK/PD trial, comparing insulin aspart with soluble human insulin was performed in elderly patients with type 2 diabetes (19 patients aged 65–83 years, mean age 70 years). The relative differences in the pharmacodynamic properties (GIR_{max} , $AUC_{GIR, 0-120 \text{ min}}$) between insulin aspart and soluble human insulin in the elderly were similar to those seen in healthy subjects and in younger patients with diabetes.

Study 001-23:

A Phase I study in healthy male subjects to compare PK/PD between Rapilin 30, NovoMix 30 has shown equivalence between the two products under fasting condition. Based on the primary PK endpoints, the 90% CI values of Geometric Least Square Mean Ratio (GLSMR) for C_{max} and AUC_{0-24} of Rapilin 30 to the NovoMix 30 are within the acceptable limits of 80.00 to 125.00%. Results of secondary PK endpoints further supported PK equivalence of Rapilin 30 to the NovoMix 30. Based on the primary PD endpoints $GIR-AUC_{0-24}$ and GIR_{max} , PD equivalence of Rapilin 30 versus NovoMix 30 was achieved since the 95% CIs of the respective GLSMR of ln-transformed-are within the acceptable limits of 75.00-133.00%. The results of secondary PD endpoints further supported PD equivalence of Rapilin 30 to the NovoMix 30. Based on evaluation of adverse events, clinical laboratory evaluation and vital signs examination, it was concluded that both Rapilin 30 and NovoMix 30 were well tolerated and were found to be safe.

Study 2011L01848:

This 24-week clinical study indicates that treatment with either Rapilin 30, developed and produced by Gan & Lee Pharmaceutical Co. Ltd., or NovoMix 30 can obtain good blood glucose control in patients with type 2 diabetes mellitus. With HbA_{1c} and fasting and two-hour postprandial venous blood glucose levels taken as criteria for judgment, treatment with Rapilin 30 is non-inferior to NovoMix 30 in therapeutic effects, with a low incidence of hypoglycemic events and adverse events, and no difference in laboratory test (blood count, urinalysis, and serum chemistry) abnormalities or electrocardiogram abnormalities between the two groups. The results of the study prove that Rapilin 30 twice daily to treat diabetes mellitus is safe, effective, and well-tolerated.

PHARMACOKINETICS

In insulin aspart, substitution of amino acid proline with aspartic acid at position B28 reduces the tendency to form hexamers as observed with human insulin. The insulin aspart in the soluble phase of insulin aspart mix 30/70 comprises 30% of the total insulin; this is absorbed more rapidly from the subcutaneous layer than the soluble insulin component of biphasic human insulin. The remaining 70% is in crystalline form as protamine-crystallised insulin aspart; this has a prolonged absorption profile similar to human NPH insulin.

The maximum serum insulin concentration is, on average, 50% higher with insulin aspart mix 30/70 than with biphasic human insulin 30. The time to maximum concentration is, on average, half of that for biphasic human insulin 30.

In healthy volunteers, a mean maximum serum concentration of 140 ± 32 pmol/L was reached about 60 minutes after a subcutaneous dose of 0.20 unit/kg body weight. The mean half-life ($t_{1/2}$) of insulin aspart mix 30/70, reflecting the absorption rate of the protamine bound fraction, was about 8-9 hours. Serum insulin levels returned to baseline 15-18 hours after a subcutaneous dose. In type 2 diabetic patients, the maximum concentration was reached about 95 minutes after dosing, and concentrations well above zero for not less than 14 hours post-dosing were measured.

Elderly: The pharmacokinetic properties of insulin aspart mix 30/70 have not been investigated in the elderly patients. However, the relative differences in pharmacokinetic properties between insulin aspart and soluble human insulin in elderly patients (65–83 years, mean age 70 years) with type 2 diabetes, were similar to those observed in healthy subjects and in younger patients with diabetes. A decreased absorption rate was observed in elderly patients, resulting in a later t_{max} (82 (interquartile range: 60–120) minutes), whereas C_{max} was similar to that observed in younger patients with type 2 diabetes and slightly lower than in patients with type 1 diabetes.

Renal and hepatic impairment: The pharmacokinetics of insulin aspart mix 30/70 has not been investigated in patients with renal or hepatic impairment.

Paediatric population: The pharmacokinetics of insulin aspart mix 30/70 has not been investigated in children or adolescents. However, the pharmacokinetic and pharmacodynamic properties of soluble insulin aspart have been investigated in children (6–12 years) and adolescents (13–17 years) with type 1 diabetes. Insulin aspart was rapidly absorbed in both age groups, with similar t_{max} as in adults. However, C_{max} differed between the age groups, stressing the importance of the individual titration of insulin aspart.

INDICATION

Rapilin 30 is indicated for treatment of patients with diabetes mellitus requiring insulin.

DOSAGE AND ADMINISTRATION

Posology

Insulin aspart mix 30/70 dosing is individual and determined in accordance with the needs of the patient. Blood glucose monitoring and insulin dose adjustments are recommended to achieve optimal glycaemic control.

In patients with type 2 diabetes, insulin aspart mix 30/70 can be given as monotherapy. Insulin aspart mix 30/70 can also be given in combination with oral antidiabetic drugs and/or GLP-1 receptor agonists.

How to start

Insulin naïve patients: For patients with type 2 diabetes, the recommended starting dose of insulin aspart mix 30/70 is 6 U at breakfast and 6 U at dinner (evening meal). However, it can also be initiated once daily with 12 U at dinner (evening meal).

How to switch

When transferring a patient from biphasic human insulin to insulin aspart mix 30/70, start with the same dose and regimen. Then titrate according to individual needs (see *the following titration guideline*). As with all insulin products, close glucose monitoring is recommended during the transfer and in the initial weeks thereafter (see *Transfer from other insulin products*).

How to intensify

Insulin aspart mix 30/70 can also be intensified from once daily to twice daily. When using insulin aspart mix 30/70 once daily, it is generally recommended to move to twice daily when reaching 30 units by splitting the dose into equal breakfast and dinner doses (50:50).

From insulin aspart mix 30/70 twice daily to thrice daily: the morning dose can be split into morning and lunchtime doses (thrice daily dosing).

How to adjust the dose

- Adjust the dose of insulin aspart mix 30/70 on the basis of the lowest pre-meal blood glucose level from the three previous days.
- Always change the mealtime dose preceding the measurement.
- Dose adjustment can be made once a week until target HbA_{1c} is reached.
- The dose should not be increased if hypoglycaemia occurred within these days.
- Adjustment of dosage may be necessary if patients undertake increased physical activity, change their usual diet or during concomitant illness.

The following titration guideline is recommended for dose adjustments:

Pre-meal blood glucose level		Insulin Aspart Mix 30/70 dose adjustment
<4.4 mmol/L	<80 mg/dL	-2 U
4.4–6.1 mmol/L	80–110 mg/dL	0
6.2–7.8 mmol/L	111–140 mg/dL	+2 U
7.9–10 mmol/L	141–180 mg/dL	+4 U
>10 mmol/L	>180 mg/dL	+6 U

In patients with type 2 diabetes, a dose reduction of 20% is recommended for patients with an HbA_{1c} less than 8% when a GLP-1 receptor agonist is added to insulin aspart mix 30/70, to minimize the risk of hypoglycaemia. For patients with an HbA_{1c} higher than 8% a dose reduction should be considered. Subsequently, dosage should be adjusted individually.

Special populations

As with all insulin products, in special populations, glucose monitoring should be intensified and the insulin aspart dosage adjusted on an individual basis.

Elderly:

Insulin aspart mix 30/70 can be used in elderly patients; however, there is limited experience with the use of insulin aspart mix 30/70 in combination with OADs in patients older than 75 years.

Renal and hepatic impairment

Renal or hepatic impairment may reduce the patient's insulin requirements.

Paediatric population

Insulin aspart mix 30/70 can be used in children and adolescents aged 10 years and above when premixed insulin is preferred. Limited clinical data exists for children aged 6 – 9 years (see Pharmacodynamic). No data are available for insulin aspart mix 30/70 in children below 6 years of age.

Method of administration

Insulin aspart mix 30/70 is for subcutaneous administration only. Insulin aspart mix 30/70 must not be administered intravenously as it may result in severe hypoglycaemia. Intramuscular administration should be avoided. Insulin aspart mix 30/70 is not to be used in insulin infusion pumps.

Insulin aspart mix 30/70 is administered subcutaneously by injection in the thigh or in the abdominal wall. If convenient, the gluteal or deltoid region may be used. Injection sites should always be rotated within the same region in order to reduce the risk of lipodystrophy and cutaneous amyloidosis (see *Warnings and precautions and Adverse reactions*). As with all insulin products, the duration of action will vary according to the dose, injection site, blood flow, temperature and level of physical activity.

Insulin aspart mix 30/70 has a faster onset of action than biphasic human insulin and should generally be given immediately before a meal. When necessary, insulin aspart mix 30/70 can be given soon after a meal.

Instruction for Use

Please bring Rapilin 30 to room temperature before use and follow the injection steps described below:

1. Check the insulin to make sure it is correct type of insulin. The insulin should appear as white suspension.
2. Please refer to the relevant reusable pen package insert, when using Rapilin 30 in conjunction with a reusable pen.
3. Rapilin 30 is injected under the skin (subcutaneously) of the upper arm, thigh, or stomach area (abdomen). Wipe the skin with an alcohol swab to clean the injection site. Let the injection site dry before injection.
4. Pinch the skin of the injection site with fingers and puncture the needle. After the pusher is pushed to the end, keep the needle under the skin for a few seconds to ensure that the correct dose is injected, then pull out the needle, and gently press the injection site with the sterile cotton ball for a few seconds, but do not rub the injection site to avoid damage to the subcutaneous tissue or cause the exudation of Rapilin 30.

CONTRAINDICATIONS

Hypersensitivity to insulin aspart or any of the excipients.

WARNING AND PRECAUTIONS

Before travelling between different time zones, the patient should seek the doctor's advice since this may mean that the patient has to take the insulin and meals at different times.

Hyperglycaemia

Inadequate dosing or discontinuation of treatment, especially in type 1 diabetes, may lead to hyperglycaemia and diabetic ketoacidosis. Usually, the first symptoms of hyperglycaemia develop gradually over a period of hours or days. They include thirst, increased frequency of urination, nausea, vomiting, drowsiness, flushed dry skin, dry mouth, loss of appetite as well as acetone odour of breath. In type 1 diabetes, untreated hyperglycaemic events eventually lead to diabetic ketoacidosis, which is potentially lethal.

Hypoglycaemia

Omission of a meal or unplanned strenuous physical exercise may lead to hypoglycaemia.

Hypoglycaemia may occur if the insulin dose is too high in relation to the insulin requirement. In case of hypoglycaemia or if hypoglycaemia is suspected, insulin aspart mix 30/70 must not be injected. After stabilization of the patient's blood glucose, adjustment of the dose should be considered.

Compared with biphasic human insulin, insulin aspart mix 30/70 may have a more pronounced glucose lowering effect up to 6 hours after injection. This may have to be compensated for in the individual patient through adjustment of insulin dose and/or food intake.

Patients whose blood glucose control is greatly improved, e.g. by intensified insulin therapy, may experience a change in their usual warning symptoms of hypoglycaemia and should be advised accordingly. Usual warning symptoms may disappear in patients with longstanding diabetes.

Tighter control of glucose levels can increase the potential for hypoglycaemic episodes and therefore require special attention during dose intensification as outlined in *Posology*.

Since insulin aspart mix 30/70 should be administered in immediate relation to a meal, the rapid onset of action should be considered in patients with concomitant diseases or treatment where a delayed absorption of food might be expected.

Concomitant illness, especially infections and feverish conditions, usually increases the patient's insulin requirements. Concomitant diseases in the kidney, liver or affecting the adrenal, pituitary or thyroid gland can require changes in the insulin dose.

When patients are transferred between different types of insulin medicinal products, the early warning symptoms of hypoglycaemia may change or become less pronounced than those experienced with their previous insulin.

Transfer from other insulin products

Transferring a patient to another type or brand of insulin should be done under strict medical supervision. Changes in strength, brand (manufacturer), type, origin (animal insulin, human insulin or insulin analogue) and/or method of manufacture (recombinant DNA versus animal source insulin) may result in the need for a change in dose. Patients transferred to insulin aspart mix 30/70 from another type of insulin may require an increased number of daily injections or a change in dose from that used with their usual insulin medicinal products. If an adjustment is needed, it may occur with the first dose or during the first few weeks or months.

Injection site reactions

As with any insulin therapy, injection site reactions may occur and include pain, redness, hives, inflammation, bruising, swelling and itching. Continuous rotation of the injection site within a given area reduces the risk of developing these reactions. Reactions usually resolve in a few days to a few weeks. On rare occasions, injection site reactions may require discontinuation of insulin aspart mix 30/70

Skin and subcutaneous tissue disorders

Patients must be instructed to perform continuous rotation of the injection site to reduce the risk of developing lipodystrophy and cutaneous amyloidosis. There is a potential risk of delayed insulin absorption and worsened glycaemic control following insulin injections at sites with these reactions. A sudden change in the injection site to an unaffected area has been reported to result in hypoglycaemia. Blood glucose monitoring is recommended after the change in the injection site from an affected to an unaffected area, and dose adjustment of antidiabetic medications may be considered.

Combination of thiazolidinediones and insulin medicinal products

Cases of congestive heart failure have been reported when thiazolidinediones was used in combination with insulin, especially in patients with risk factor for development of congestive heart failure. This should be kept in mind if treatment with the combination of thiazolidinediones and insulin medicinal products is considered. If the combination is used, patients should be observed for signs and symptoms of congestive heart failure, weight gain and oedema. Thiazolidinediones should be discontinued if any deterioration in cardiac symptoms occurs.

Avoidance of accidental mix-ups/medication errors

Patients must be instructed to always check the insulin label before each injection to avoid accidental mix-ups between insulin aspart mix 30/70 and other insulin products.

Insulin Antibodies

Insulin administration may cause insulin antibodies to form. In rare cases, the presence of such insulin antibodies may necessitate adjustment of the insulin dose in order to correct a tendency to hyper- or hypoglycaemia.

INTERACTIONS WITH OTHER MEDICAMENTS

A number of medicinal products are known to interact with the glucose metabolism.

The following substances may reduce the patient's insulin requirements:

Oral antidiabetic medicinal products, monoamine oxidase inhibitors (MAOI), beta-blockers, angiotensin converting enzyme (ACE) inhibitors, salicylates, anabolic steroids and sulfonamides.

The following substances may increase the patient's insulin requirements:

Oral contraceptives, thiazides, glucocorticoids, thyroid hormones, sympathomimetics, growth hormone and danazol.

Beta-blockers may mask the symptoms of hypoglycaemia.

Octreotide/lanreotide may either increase or decrease the insulin requirement.

Alcohol may intensify or reduce the hypoglycaemic effect of insulin.

PREGNANCY AND LACTATION

Pregnancy

There is limited clinical experience with insulin aspart mix 30/70 in pregnancy. Insulin aspart mix 30/70 has not been investigated in pregnant women. However, data from two randomised controlled clinical do not indicate any adverse effect of insulin aspart on pregnancy or on the health of the foetus/newborn when compared to soluble human insulin.

In general, intensified blood glucose control and monitoring of pregnant women with diabetes are recommended throughout pregnancy and when contemplating pregnancy. Insulin requirements usually fall in the first trimester and increase subsequently during the second and third trimesters. After delivery, insulin requirements return rapidly to pre-pregnancy levels.

Lactation

There are no restrictions on treatment with insulin aspart mix 30/70 during breast-feeding. Insulin treatment of the nursing mother presents no risk to the baby. However, the insulin aspart mix 30/70 dose may need to be adjusted.

ADVERSE REACTIONS

a. Summary of the safety profile

Adverse reactions observed in patients using insulin aspart mix 30/70 are mainly due to the pharmacologic effect of insulin. The most frequently reported adverse reaction during treatment is hypoglycaemia. The frequencies of hypoglycaemia vary with patient population, dose regimens and level of glycaemic control (please see below).

At the beginning of the insulin treatment, refraction anomalies, oedema and injection site reactions (pain, redness, hives, inflammation, bruising, swelling and itching at the injection site) may occur. These reactions are usually of a transitory nature. Fast improvement in blood glucose control maybe associated with acute painful neuropathy, which is usually reversible. Intensification of insulin therapy with abrupt improvement in glycaemic control may be associated with temporary worsening of diabetic retinopathy, while long-term improved glycaemic control decreases the risk of progression of diabetic retinopathy.

b. Tabulated list of adverse reactions

The adverse reactions listed below are based on clinical trial data and classified according to MedDRA frequency and System Organ Class. Frequency categories are defined according to the following convention: Very common ($\geq 1/10$); common ($\geq 1/100$ to $< 1/10$); uncommon ($\geq 1/1,000$ to $< 1/100$); rare ($\geq 1/10,000$ to $< 1/1,000$); very rare ($< 1/10,000$); not known (cannot be estimated from the available data).

Nervous system disorders	Rare – Peripheral neuropathy (painful neuropathy)
Eye disorders	Uncommon – Refraction disorders
	Uncommon – Diabetic retinopathy
Skin and subcutaneous tissue disorders	Uncommon – Lipodystrophy*
	Not known – Cutaneous amyloidosis*†
General disorders and administration site conditions	Uncommon – Oedema
	Uncommon – Injection site reactions
Immune system disorders	Uncommon – Urticaria, rash, eruptions
	Very rare – Anaphylactic reactions*
Metabolism and nutrition disorders	Very common – Hypoglycaemia*

* see Description of selected adverse reactions.

† ADR from postmarketing sources.

c. Description of selected adverse reactions

Anaphylactic reactions

The occurrence of generalised hypersensitivity reactions (including generalised skin rash, itching, sweating, gastrointestinal upset, angioneurotic oedema, difficulties in breathing, palpitation and reduction in blood pressure) is very rare but can potentially be life-threatening.

Hypoglycaemia

The most frequently reported adverse reaction is hypoglycaemia. It may occur if the insulin dose is too high in relation to the insulin requirement. Severe hypoglycaemia may lead to unconsciousness and/or convulsions and may result in temporary or permanent impairment of brain function or even death. The symptoms of hypoglycaemia usually occur suddenly. They may include cold sweats, cool pale skin, fatigue, nervousness or tremor, anxiousness, unusual tiredness or weakness, confusion, difficulty in concentrating, drowsiness, excessive hunger, vision changes, headache, nausea and palpitation.

In clinical trials, the frequency of hypoglycaemia varied with patient population, dose regimens and level of glycaemic control. During clinical trials, the overall rates of hypoglycaemia did not differ between patients treated with insulin aspart compared to human insulin.

Skin and subcutaneous tissue disorders

Lipodystrophy (including lipohypertrophy, lipoatrophy) and cutaneous amyloidosis may occur at the injection site and delay local insulin absorption. Continuous rotation of the injection site within the given injection area may help to reduce or prevent these reactions.

SYMPTOMS AND TREATMENT OF OVERDOSE

A specific overdose for insulin cannot be defined, however, hypoglycaemia may develop over sequential stages if too high doses relative to the patient's requirement are administered:

- Mild hypoglycaemic episodes can be treated by oral administration of glucose or sugary products. It is therefore recommended that the diabetic patient always carries sugar-containing products.
- Severe hypoglycaemic episodes, where the patient has become unconscious, can be treated with glucagon (0.5 to 1 mg) given intramuscularly or subcutaneously by a trained person, or with glucose given intravenously by a healthcare professional. Glucose must be given intravenously, if the patient does not respond to glucagon within 10 to 15 minutes. Upon regaining consciousness, administration of oral carbohydrates is recommended for the patient in order to prevent a relapse.

EFFECT ON ABILITY TO DRIVE AND USE MACHINES

The patient's ability to concentrate and react may be impaired as a result of hypoglycaemia. This may constitute a risk in situations where these abilities are of special importance (e.g. driving a car or operating machinery).

Patients should be advised to take precautions to avoid hypoglycaemia while driving or operating a machine. This is particularly important in those who have reduced or absent awareness of the warning signs of hypoglycaemia or have frequent episodes of hypoglycaemia. The advisability of driving or operating a machine should be considered in these circumstances.

PRECLINICAL SAFETY DATA

Non-clinical data reveal no special hazard for humans based on conventional studies of safety pharmacology, repeated dose toxicity, genotoxicity and toxicity to reproduction.

In *in vitro* tests, including binding to insulin and IGF-1 receptor sites and effects on cell growth, insulin aspart behaved in a manner that closely resembled human insulin. Studies also demonstrate that the dissociation of binding to the insulin receptor of insulin aspart is equivalent to human insulin.

INCOMPATIBILITIES

In the absence of compatibility studies, this medicinal product must not be mixed with other medicinal products.

STORAGE CONDITIONS

Unopened cartridges

Store in a refrigerator (2°C -8°C). Do not freeze.

Keep away from the cooling element.

Keep the cartridge in the outer carton in order to protect from light.

In-use cartridges

Do not refrigerate. Store below 30°C. Use within 4 weeks.

Do not freeze. Do not expose from excessive heat or direct light.

SHELF LIFE

Unopened cartridges

24 months

In-use cartridges

28 days

DOSAGE FORM AND PACKAGING AVAILABLE

3 mL suspension in a cartridge (Type I colourless glass) with a bromobutyl plunger and a cap comprising a rubber liner (rubber and an aluminum seal). The cartridge contains a glass bead to facilitate resuspension. 1 cartridge/box.

MANUFACTURER**Gan & Lee Pharmaceuticals**

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PRODUCT REGISTRATION HOLDER**GoodScience Sdn Bhd**

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